```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

df=pd.read_csv('BigBasket Products.csv')

df.head(12)
```

```
    index                                            product  \
0       1             Garlic Oil - Vegetarian Capsule 500 mg
1       2                               Water Bottle - Orange
2       3                      Brass Angle Deep - Plain, No.2
3       4  Cereal Flip Lid Container/Storage Jar - Assort...
4       5                  Creme Soft Soap - For Hands & Body
5       6                   Germ - Removal Multipurpose Wipes
6       7                                        Multani Mati
7       8                   Hand Sanitizer - 70% Alcohol Base
8       9  Biotin & Collagen Volumizing Hair Shampoo + Bi...
9      10                Scrub Pad - Anti- Bacterial, Regular
10     11                           Wheat Grass Powder - Raw
11     12                      Butter Cookies Gold Collection

                  category            sub_category              brand
\
0         Beauty & Hygiene               Hair Care  Sri Sri Ayurveda

1   Kitchen, Garden & Pets   Storage & Accessories        Mastercook

2     Cleaning & Household             Pooja Needs               Trm

3     Cleaning & Household    Bins & Bathroom Ware            Nakoda

4         Beauty & Hygiene        Bath & Hand Wash             Nivea

5     Cleaning & Household    All Purpose Cleaners     Nature Protect

6         Beauty & Hygiene               Skin Care         Satinance

7         Beauty & Hygiene        Bath & Hand Wash           Bionova

8         Beauty & Hygiene               Hair Care        StBotanica

9     Cleaning & Household  Mops, Brushes & Scrubs       Scotch brite

10    Gourmet & World Food  Cooking & Baking Needs         NUTRASHIL

11    Gourmet & World Food   Chocolates & Biscuits          Sapphire
```

```
    sale_price  market_price                           type  rating  \
0        220.0         220.0                Hair Oil & Serum     4.1
1        180.0         180.0          Water & Fridge Bottles     2.3
2        119.0         250.0                 Lamp & Lamp Oil     3.4
3        149.0         176.0       Laundry, Storage Baskets     3.7
4        162.0         162.0            Bathing Bars & Soaps     4.4
5        169.0         199.0  Disinfectant Spray & Cleaners     3.3
6         58.0          58.0                      Face Care     3.6
7        250.0         250.0          Hand Wash & Sanitizers     4.0
8       1098.0        1098.0           Shampoo & Conditioner     3.5
9         20.0          20.0       Utensil Scrub-Pad, Glove     4.3
10       261.0         290.0             Flours & Pre-Mixes     4.0
11       600.0         600.0       Luxury Chocolates, Gifts     2.2

                                          description
0   This Product contains Garlic Oil that is known...
1   Each product is microwave safe (without lid), ...
2   A perfect gift for all occasions, be it your m...
3   Multipurpose container with an attractive desi...
4   Nivea Creme Soft Soap gives your skin the best...
5   Stay protected from contamination with Multipu...
6   Satinance multani matti is an excellent skin t...
7   70%Alcohol based is gentle of hand leaves skin...
8   An exclusive blend with Vitamin B7 Biotin, Hyd...
9   Scotch Brite Anti- Bacterial Scrub Pad thoroug...
10  Wheatgrass is a superfood potent health food w...
11  Enjoy a tin full of delicious butter cookies m...
```

```
df.shape
```

```
(27555, 10)
```

There are 27555 rows and 10 columns in the dataset

```
df.columns
```

```
Index(['index', 'product', 'category', 'sub_category', 'brand',
'sale_price',
       'market_price', 'type', 'rating', 'description'],
      dtype='object')
```

Property Characteristics

- **index**: Unique identifier assigned to each product entry
- **product**: Name/title of the product listed on BigBasket
- **category**: Main category under which the product is classified

- **sub_category**: More specific classification within the main category
- **brand**: Brand name of the product
- **sale_price**: Discounted price at which the product is being sold
- **market_price**: Original price of the product before discount
- **type**: Type or variant of the product based on its characteristics
- **rating**: Customer rating given to the product
- **description**: Detailed textual description of the product

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   index         27555 non-null  int64
 1   product       27554 non-null  object
 2   category      27555 non-null  object
 3   sub_category  27555 non-null  object
 4   brand         27554 non-null  object
 5   sale_price    27549 non-null  float64
 6   market_price  27555 non-null  float64
 7   type          27555 non-null  object
 8   rating        18919 non-null  float64
 9   description   27440 non-null  object
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB

df.describe()

             index     sale_price  market_price        rating
count  27555.00000   27549.000000  27555.000000  18919.000000
mean   13778.00000     334.648391    382.056664      3.943295
std     7954.58767    1202.102113    581.730717      0.739217
min        1.00000       2.450000      3.000000      1.000000
25%     6889.50000      95.000000    100.000000      3.700000
50%    13778.00000     190.320000    220.000000      4.100000
75%    20666.50000     359.000000    425.000000      4.300000
max    27555.00000  112475.000000  12500.000000      5.000000
```

Observations from Descriptive Statistics

- The dataset contains 27,555 records, but ratings are missing for a significant number of products
- The average sale price (₹334) is lower than the average market price (₹382), indicating discounts on most products
- There is a huge variation in sale price (std ~1202), suggesting presence of extreme values/outliers
- The maximum sale price (₹112,475) is unusually high compared to the 75th percentile (~₹359), confirming strong outliers
- Ratings are generally high with an average of ~3.94, indicating overall positive customer feedback
- The rating distribution is slightly skewed towards higher values (median ~4.1)
- Price-related features show a right-skewed distribution due to high-value products

```
#duplicacy
print("no of duplicate enteries",df.duplicated().sum())
```

```
no of duplicate enteries 0
```

- The dataset contains no duplicate entries, ensuring data quality and eliminating the need for duplicate handling

```
df.isnull().sum()
```

```
index              0
product            1
category           0
sub_category       0
brand              1
sale_price         6
market_price       0
type               0
rating          8636
description      115
dtype: int64
```

```
df.isnull().sum()[df.isnull().sum()>0]
```

```
product           1
brand             1
sale_price        6
rating         8636
```

```
description     115
dtype: int64
```

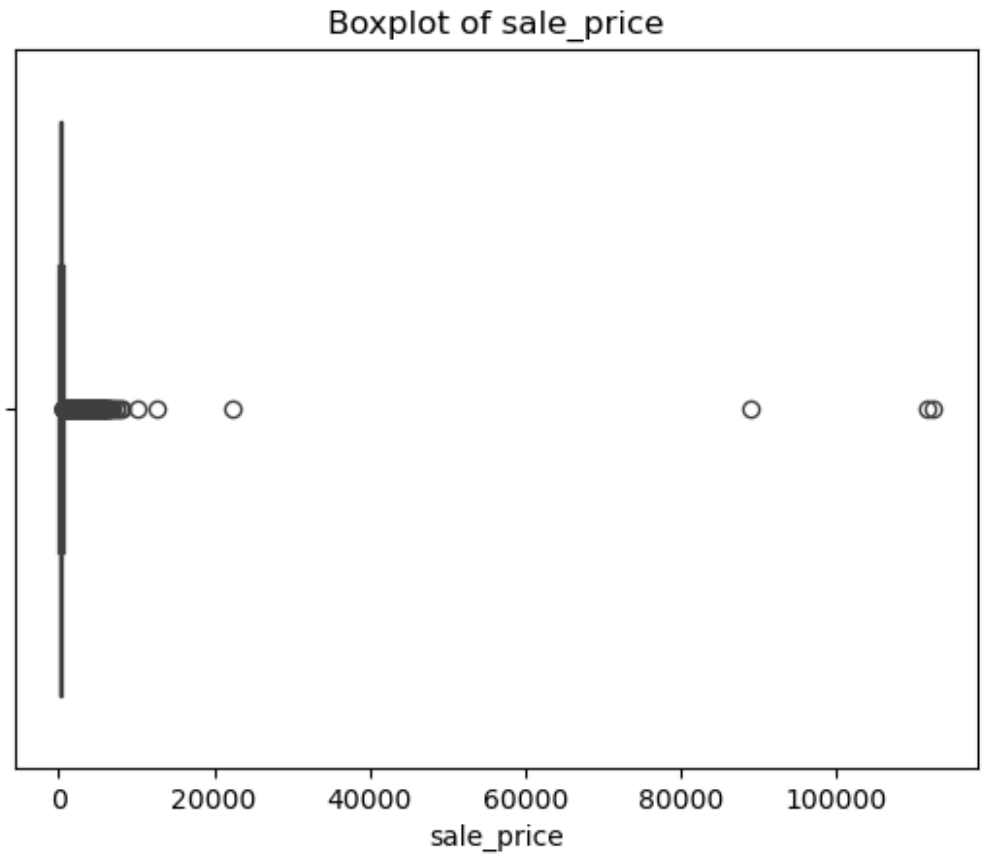
```
df = df.dropna(subset=['product'])

df['brand'] = df['brand'].fillna('Unknown')

df['sale_price'] = df['sale_price'].fillna(df['sale_price'].median())

df['rating'] = df['rating'].fillna(df['rating'].median())

df['description'] = df['description'].fillna('No description
available')

df.isnull().sum()
```

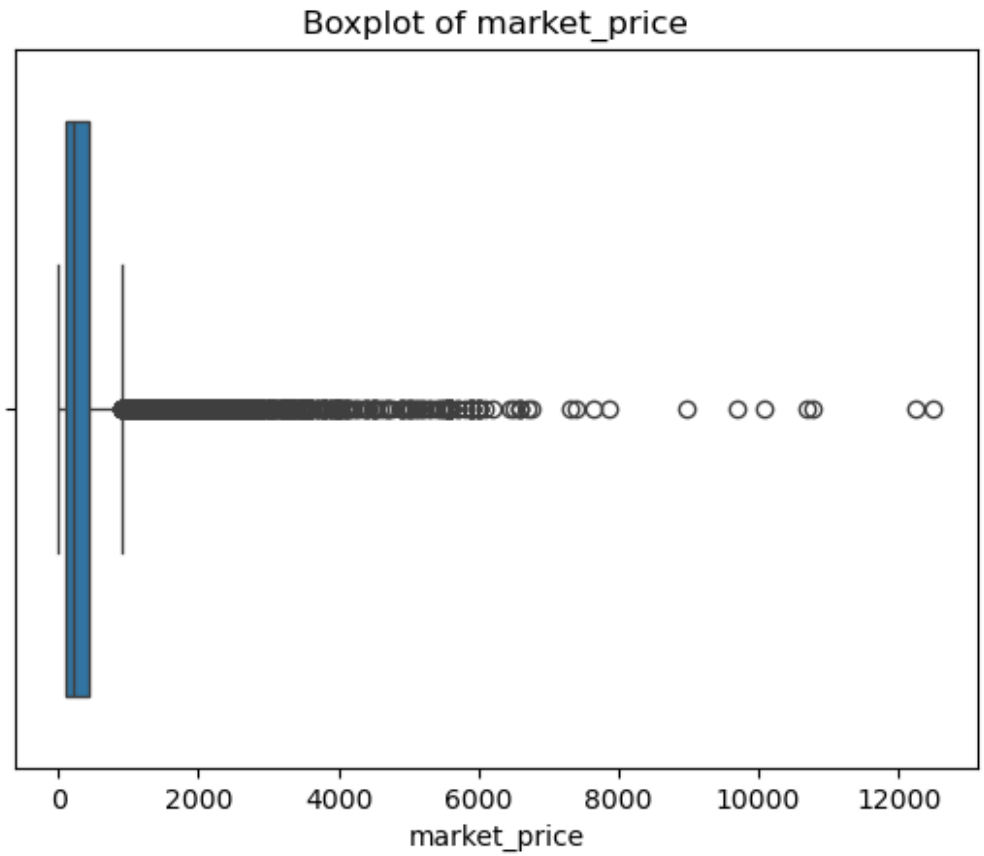
```
index           0
product         0
category        0
sub_category    0
brand           0
sale_price      0
market_price    0
type            0
rating          0
description     0
dtype: int64
```

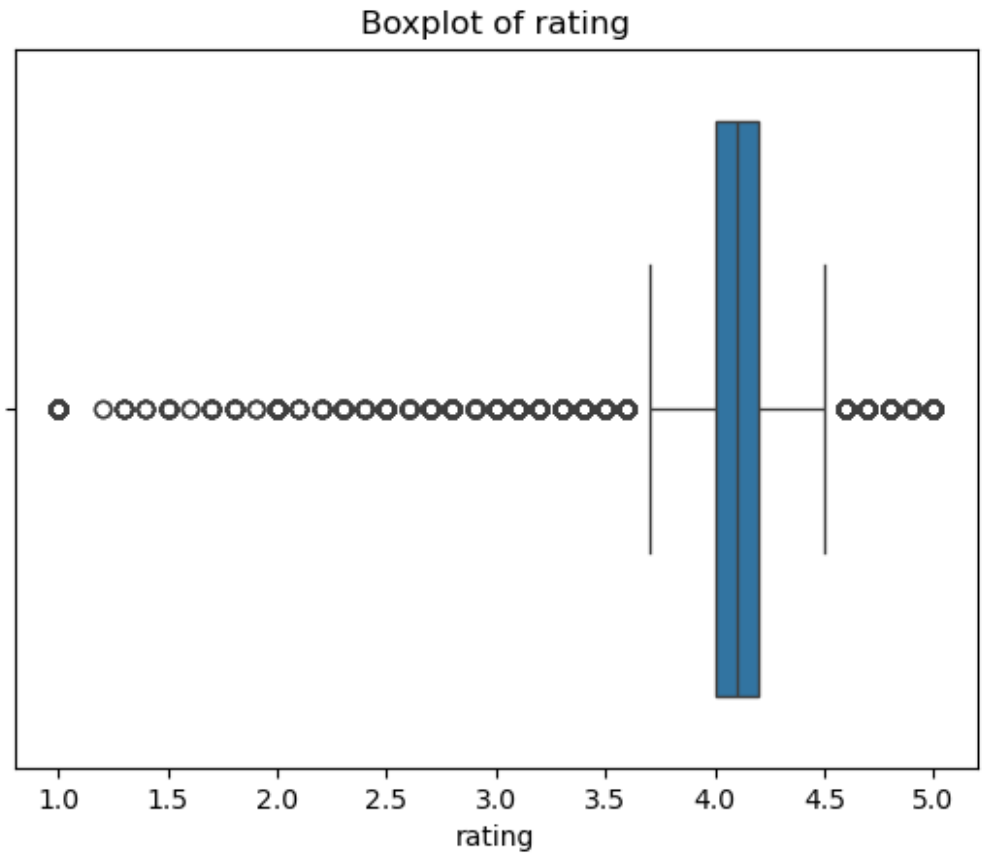
- Product column had 1 missing value, which was removed as it is a key identifier
- Brand column missing values were replaced with "Unknown" to retain data
- Sale price missing values were imputed using median due to presence of outliers
- Rating had a large number of missing values (~8636), filled using median to avoid data loss
- Description missing values were replaced with a placeholder text

Outlier Detection

```
num_cols = ['sale_price', 'market_price', 'rating']
for col in num_cols:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
    plt.show()
```


Boxplot of sale_price
0
20000
40000
60000
80000
100000
sale_price


Boxplot of market_price
0
2000
4000
6000
8000
10000
12000
market_price

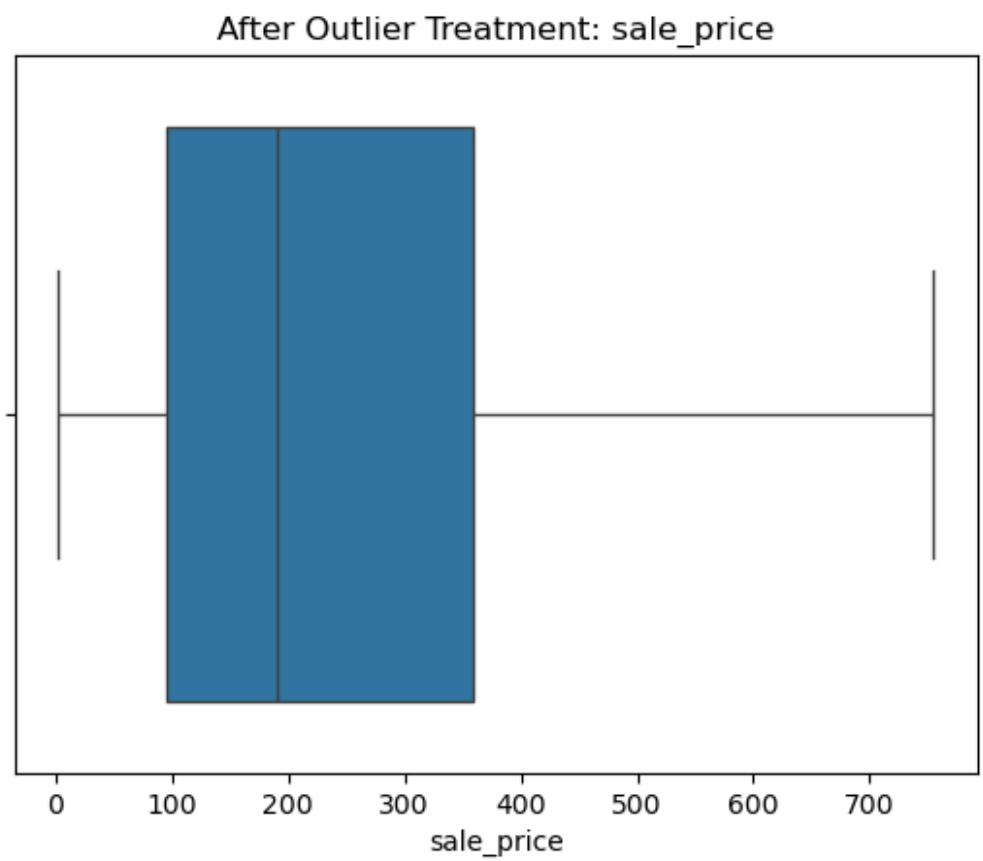


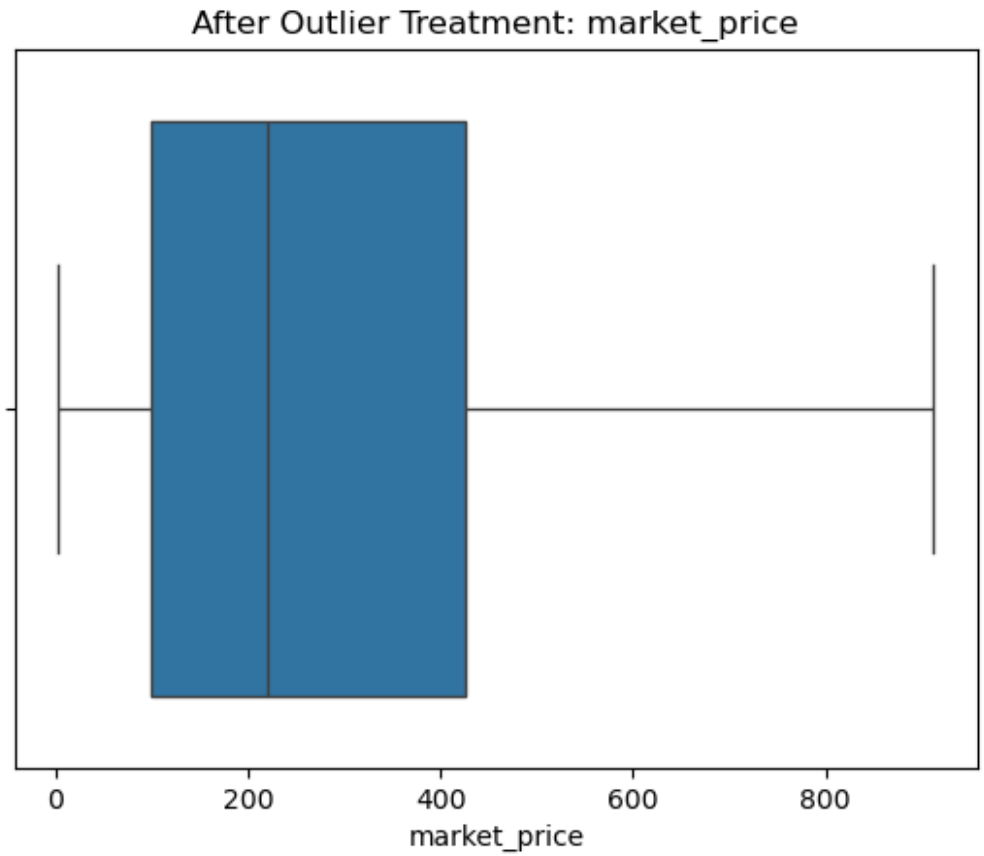

```
def detect_outliers(column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[column] < lower_bound) | (df[column] >
upper_bound)]

    print(f"{column}: {len(outliers)} outliers")

for col in num_cols:
    detect_outliers(col)
```

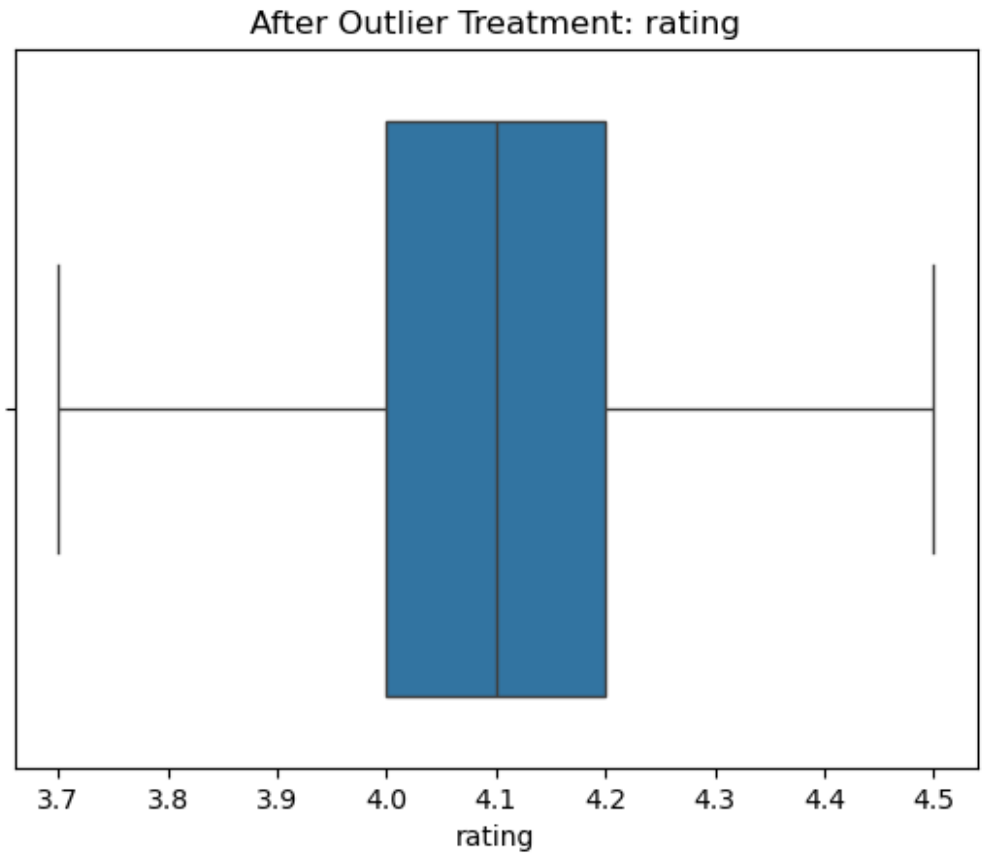
```
sale_price: 2205 outliers
market_price: 2147 outliers
rating: 6372 outliers
```

```
def cap_outliers(column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
```

```
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    df[column] = np.where(df[column] < lower_bound, lower_bound,
                          np.where(df[column] > upper_bound,
upper_bound, df[column]))

for col in num_cols:
    cap_outliers(col)

for col in num_cols:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f'After Outlier Treatment: {col}')
    plt.show()
```




After Outlier Treatment: market_price
0
200
400
600
800
market_price



Outlier Detection and Treatment

- Outliers were identified using the IQR (Interquartile Range) method
- Significant outliers were observed in price-related columns such as sale_price and market_price
- Instead of removing data, outliers were capped using lower and upper bounds
- This approach helps retain data while reducing the impact of extreme values

```
df['discount_amount'] = df['market_price'] - df['sale_price']

df['discount_percent'] = ((df['market_price'] - df['sale_price']) /
df['market_price']) * 100

df.columns

Index(['index', 'product', 'category', 'sub_category', 'brand',
'sale_price',
       'market_price', 'type', 'rating', 'description',
'discount_amount',
       'discount_percent'],
      dtype='object')
```

```
df.head(5)
```

```
   index                                                product  \
0      1             Garlic Oil - Vegetarian Capsule 500 mg
1      2                              Water Bottle - Orange
2      3                      Brass Angle Deep - Plain, No.2
3      4  Cereal Flip Lid Container/Storage Jar - Assort...
4      5                  Creme Soft Soap - For Hands & Body

                 category           sub_category              brand  \
0        Beauty & Hygiene              Hair Care  Sri Sri Ayurveda
1  Kitchen, Garden & Pets  Storage & Accessories         Mastercook
2    Cleaning & Household            Pooja Needs                Trm
3    Cleaning & Household   Bins & Bathroom Ware             Nakoda
4        Beauty & Hygiene       Bath & Hand Wash              Nivea

   sale_price  market_price                       type  rating  \
0       220.0         220.0          Hair Oil & Serum     4.1
1       180.0         180.0    Water & Fridge Bottles     3.7
2       119.0         250.0           Lamp & Lamp Oil     3.7
3       149.0         176.0  Laundry, Storage Baskets     3.7
4       162.0         162.0      Bathing Bars & Soaps     4.4

                                          description  discount_amount
\
0  This Product contains Garlic Oil that is known...              0.0

1  Each product is microwave safe (without lid), ...              0.0

2  A perfect gift for all occasions, be it your m...            131.0

3  Multipurpose container with an attractive desi...             27.0

4  Nivea Creme Soft Soap gives your skin the best...              0.0


   discount_percent
0          0.000000
1          0.000000
2         52.400000
3         15.340909
4          0.000000
```

- Created a new feature 'discount_amount' to calculate the price difference between market price and sale price
- Created 'discount_percent' to understand the percentage discount offered on products
- These features help in analyzing pricing strategies and promotional trends

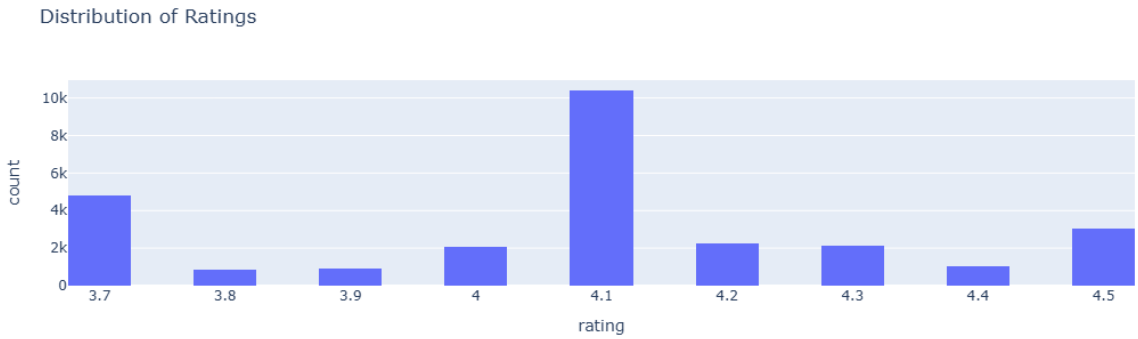
Exploratory Data Analysis

****1.Univariate****

```
#sales price
plt.figure(figsize=(12,5))
sns.histplot(df['sale_price'], bins=50, kde=True, color='skyblue')
plt.title("Distribution of Sale Price")
plt.xlabel("Sale Price")
plt.ylabel("Count")
plt.show()
```

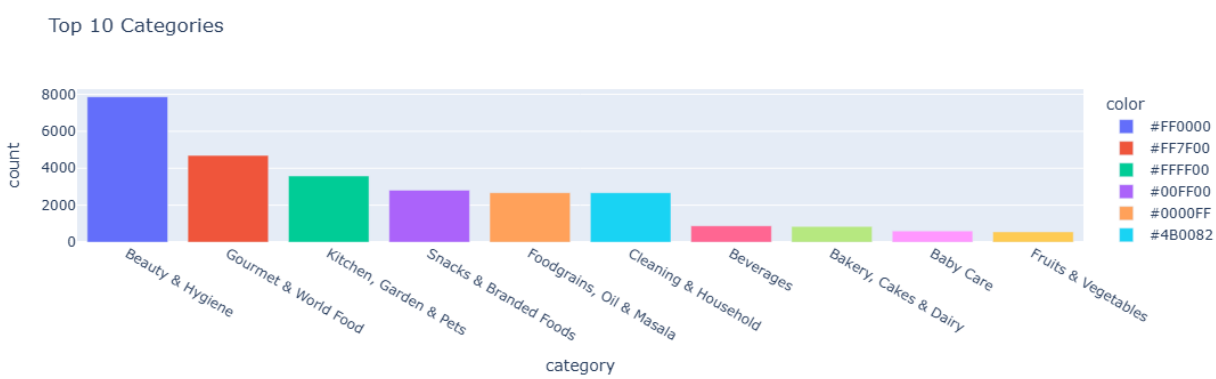
- The distribution of sale price is right-skewed, indicating the presence of higher-priced products
- Most products are concentrated in the lower price range (approximately ₹50 to ₹300)
- A gradual decline in frequency is observed as price increases
- Few products exist in the higher price range, showing a long tail distribution
- The peak around lower price bins suggests that affordable products dominate the platform

```
#ratings
fig = px.histogram(df, x='rating', nbins=30,title='Distribution of
Ratings')
fig.show()
```



- Most product ratings are concentrated between 4.0 and 4.2, indicating generally high customer satisfaction
- The highest frequency is observed around rating 4.1, making it the most common rating
- Very few products have ratings below 3.8, suggesting limited poor-performing products
- The distribution is slightly left-skewed, with more values on the higher rating side
- Overall, the platform maintains a strong rating profile across products

```
#df.columns

#cateogries
top_cat = df['category'].value_counts().head(10).reset_index()
colors=['#FF0000','#FF7F00','#FFFF00','#00FF00','#0000FF','#4B0082','#
9400D3','#FF1493','#00CED1','#FFD700']
top_cat.columns = ['category', 'count']
fig = px.bar(top_cat, x='category', y='count',color=colors,title='Top
10 Categories')
fig.show()
```



- Beauty & Hygiene is the most dominant category with the highest number of products

- Gourmet & World Food and Kitchen, Garden & Pets also contribute significantly to the product range

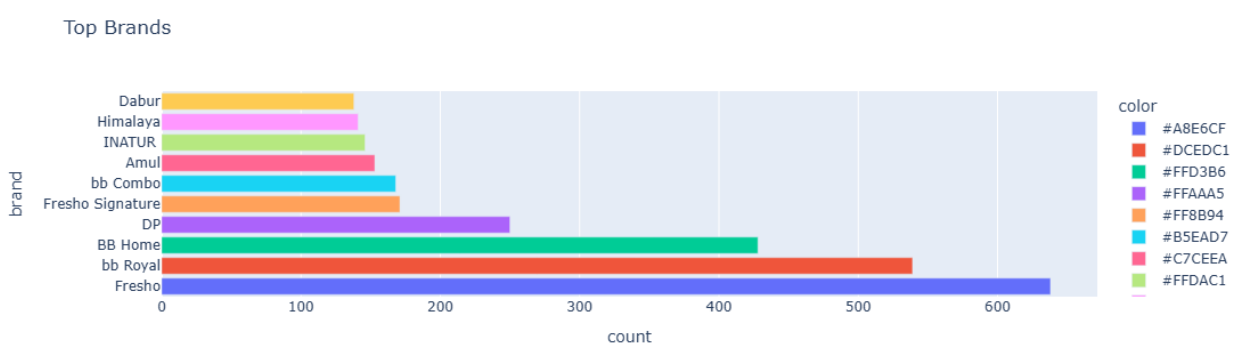
- Categories like Snacks & Branded Foods and Foodgrains, Oil & Masala show strong presence, indicating high demand for daily essentials

- Cleaning & Household has moderate representation, reflecting regular utility purchases

- Categories such as Beverages, Bakery, Baby Care, and Fruits & Vegetables have comparatively fewer listings

- The distribution suggests a stronger focus on packaged and personal care products over fresh produce

```
#brand
top_brand = df['brand'].value_counts().head(10).reset_index()
colors =
['#A8E6CF','#DCEDC1','#FFD3B6','#FFAAA5','#FF8B94','#B5EAD7','#C7CEEA'
,'#FFDAC1','#E2F0CB','#FFB7B2']
top_brand.columns = ['brand', 'count']
fig = px.bar(top_brand, x='count', y='brand',
orientation='h',color=colors,title='Top Brands')
fig.show()
```



- Fresho is the most dominant brand with the highest number of product listings

- bb Royal and BB Home also have a strong presence, indicating BigBasket's focus on in-house/private labels

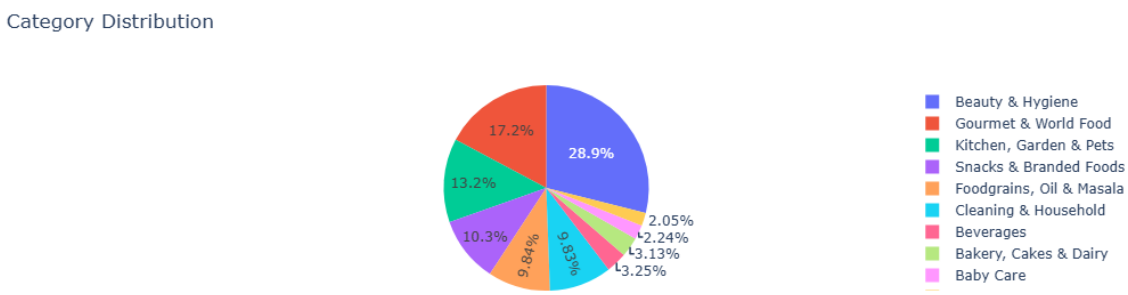
- DP shows moderate representation compared to leading brands

- Other brands like Fresho Signature, bb Combo, Amul, and INATUR have relatively lower product counts

- Established brands like Dabur and Himalaya are present but not as dominant in terms of product volume

- The distribution highlights a strong emphasis on private label brands over third-party brands

```
#category share
colors=['#A8DADC','#457B9D','#1D3557','#F1FAEE','#E63946','#FFB4A2','#6D6875','#B5838D','#84A59D','#F4A261']
fig = px.pie(top_cat, names='category', values='count',color=colors, title='Category Distribution')
fig.show()
```



- Beauty & Hygiene holds the largest share (~28.9%), making it the dominant category

- Gourmet & World Food (17.2%) and Kitchen, Garden & Pets (13.2%) also contribute significantly

- Snacks & Branded Foods and Foodgrains categories show moderate representation

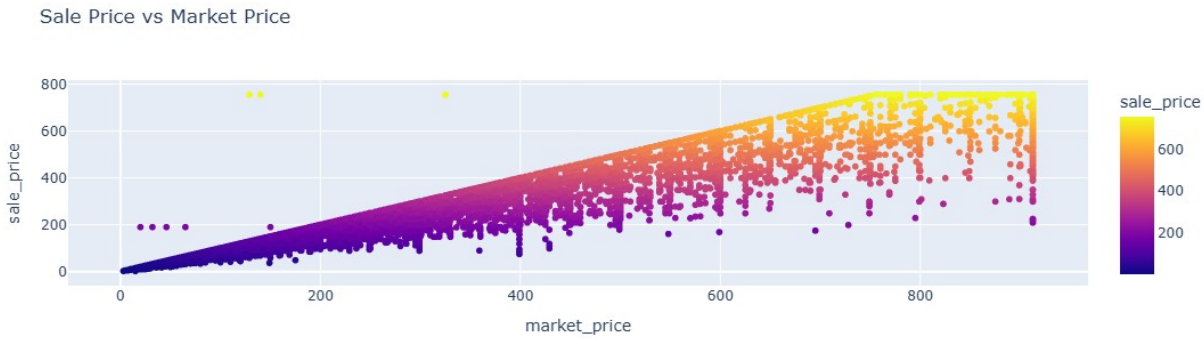
- Cleaning & Household has a noticeable but smaller share compared to top categories

- Categories like Beverages, Bakery, Baby Care, and others contribute minimal proportions

- The distribution indicates a stronger focus on personal care and packaged goods over niche categories

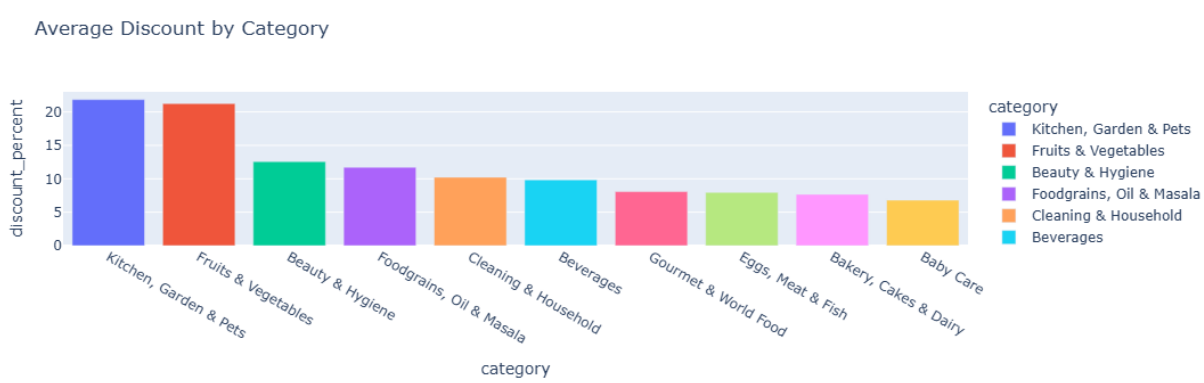
****2.Bivariate****

```
#sales price vs market price
fig = px.scatter(df, x='market_price', y='sale_price',color='sale_price',
                 title='Sale Price vs Market Price')
fig.show()
```

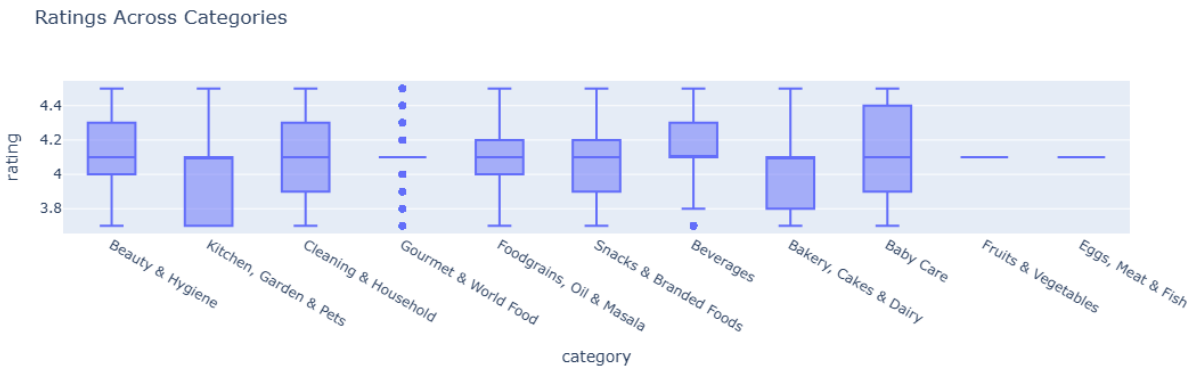


- A strong positive relationship is observed between market price and sale price
- Most data points lie below the diagonal trend, indicating that products are generally sold at a discount
- The spread increases as prices increase, suggesting higher variability in discounts for expensive products
- A few scattered points indicate possible anomalies or products with unusual pricing patterns
- The dense clustering in lower price ranges shows that most products are priced affordably

```
#discount% by category
cat_discount = df.groupby('category')
['discount_percent'].mean().sort_values(ascending=False).head(10).rese
t_index()
fig =
px.bar(cat_discount,x='category',y='discount_percent',color='category'
,title='Average Discount by Category')
fig.show()
```

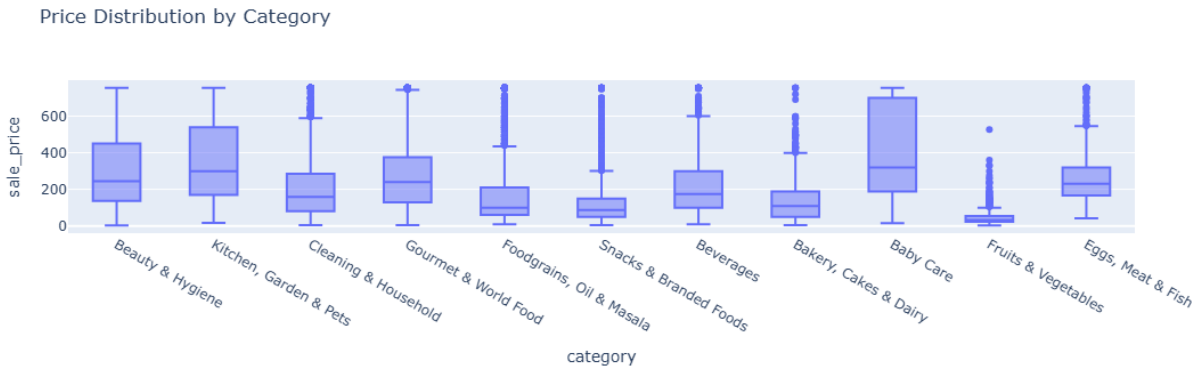


```
#rating vs discount
fig = px.box(df, x='category', y='rating',
             title='Ratings Across Categories')
fig.show()
```



- Ratings across most categories are consistently high, generally ranging between 4.0 and 4.3
- Median ratings for most categories are around 4.1, indicating stable customer satisfaction
- Categories like Beverages and Baby Care show slightly higher median ratings
- Kitchen, Garden & Pets and Bakery categories exhibit slightly wider spread, indicating variation in customer experience
- Presence of a few lower-end outliers suggests occasional poorly rated products
- Overall, rating distribution is fairly uniform across categories with no major underperforming segment

```
#sales price vs category
fig = px.box(df, x='category', y='sale_price',
             title='Price Distribution by Category')
fig.show()
```

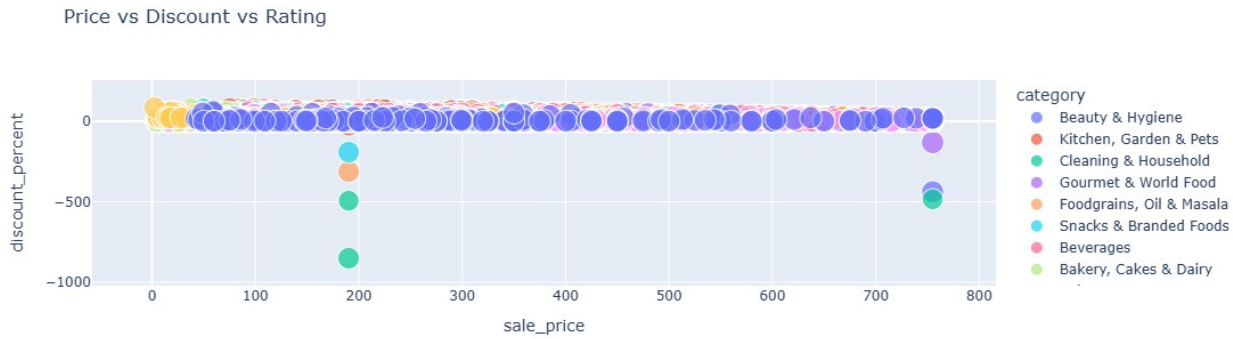


- Significant variation in sale prices is observed across different categories
- Baby Care and Kitchen, Garden & Pets show higher median prices, indicating relatively premium products
- Categories like Fruits & Vegetables and Foodgrains have lower price ranges, reflecting everyday essentials
- Wide interquartile ranges in some categories indicate diverse product pricing within the same category
- Presence of multiple outliers suggests availability of premium/high-priced items across categories
- Categories such as Snacks & Branded Foods and Bakery items have relatively lower and more concentrated price ranges

```
df.columns

Index(['index', 'product', 'category', 'sub_category', 'brand',
'sale_price',
       'market_price', 'type', 'rating', 'description',
'discount_amount',
       'discount_percent'],
      dtype='object')
```

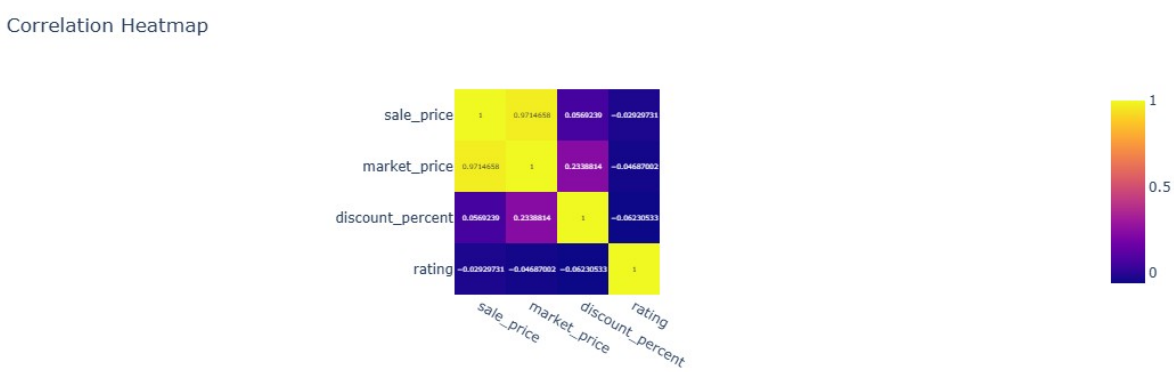
****3.Multivariate****

```
fig = px.scatter(df,
                 x='sale_price',
                 y='discount_percent',
                 size='rating',
                 color='category',
                 title='Price vs Discount vs Rating')
fig.show()
```



- Most products cluster around lower discount values, indicating moderate discounting across the platform
- Ratings remain consistently high across different price and discount levels, showing weak correlation between discount and rating
- Higher-priced products do not necessarily receive higher ratings, suggesting price is not a key driver of customer satisfaction
- A few extreme negative discount values (outliers) are visible, indicating possible data inconsistencies or pricing anomalies
- Dense clustering across categories suggests similar pricing and discount strategies across product types

```
corr = df[['sale_price', 'market_price', 'discount_percent',
'rating']].corr()

fig = px.imshow(corr, text_auto=True, title='Correlation Heatmap')
fig.show()
```



- A very strong positive correlation (~0.97) exists between sale_price and market_price, indicating consistent pricing structure

- Discount_percent shows a weak positive correlation with market_price (~0.23), suggesting higher-priced items may have slightly higher discounts
- Sale_price has very low correlation with discount_percent, indicating discounts vary independently of selling price
- Rating has negligible (near zero or slightly negative) correlation with price and discount features
- This suggests that customer ratings are not significantly influenced by price or discount levels